

## Sachcith G N

B.Tech CSE (AI), Amrita Vishwa Vidyapeetham, Coimbatore  
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### Education

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**Amrita Vishwa Vidyapeetham, Coimbatore** B.Tech CSE (AI), CGPA: **9.23/10.0** Expected 2028  
**IIT Madras BS in Data Science and Applications**, CGPA: **7.0/10.0** Expected 2028

### Publications

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#### Research Under Review (IEEE Transactions on Reliability)

- Proposed YOLOv11-based multi-modal framework for photovoltaic fault detection outperforming 36+ state-of-the-art models on PVEL-AD and PTI datasets.

#### IEEE PES Hong Kong Conference Publication

[Link](#)

- Worked on research design, implementation, experimentation, and evaluation.

### Competitive Programming

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- Active competitive programmer on Codeforces with experience in algorithms, data structures, and contest programming.
- Rating 1049

### Research Engineering Projects

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#### HMC Image Segmentation

[GitHub](#)

- Implemented EM, ICE, MPM, Hilbert-curve scanning and probabilistic image segmentation pipelines from a 2D-HMC research.
- Reproduced mathematical models in MATLAB and evaluated segmentation quality on synthetic and medical imaging datasets.

#### Embedded Audio Intelligence System

[GitHub](#)

- Built a custom audio intelligence framework from scratch in C++, implementing signal-processing pipelines, wavelet (DWT) and Fourier-domain (STFT) feature extraction, statistical descriptors, and neural-network inference without relying on high-level ML frameworks.
- Deployed a compressed MLP-based classifier on ESP32 by translating a Python training pipeline into Embedded C++, achieving real-time inference through memory-aware optimization and custom numerical computing primitives.

#### ERP Management System

- Built a modular ERP system comprising inventory, stock, transaction, and entity-management modules with secure JWT-based authentication and real-time client-server synchronization.
- Developed a transaction-processing engine that maintains financial consistency through automated balance updates, stock tracking, historical ledger correction, and business-rule enforcement.

#### Othello AI Engine

[GitHub](#)

- Built a complete Othello engine from scratch with legal move generation, state management, and adversarial game mechanics.
- Implemented Alpha-Beta Pruned Minimax search with custom positional heuristics and investigated neural-network-based evaluation through a YOLO-driven board-state classification pipeline.

### Technical Skills

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**Programming Languages:** Python, C++, C, Java, MATLAB, SQL

**Areas:** Systems Programming, Embedded Systems, Machine Learning, Computer Vision, Signal Processing, Algorithms, Research Engineering

**Frameworks & Tools:** Flask, Socketio, Linux, Git, ESP32